

ANALYSIS_STATUS: SUCCEEDED

Glial Inflammatory and ER-Stress Model of Depression in Alzheimer Disease — Computational Replication Report

Disease: Alzheimer Disease · **Hypothesis ID:** `glial_depression_in_ad_model` **Source dataset:** He et al. 2026, *Nature Medicine* — Supplementary Data zip (`41591_2025_4128_MOESM3_ESM.zip`), **PMID:** [42778763](#), doi:10.1038/s41591-025-04128-1 **Data lineage:** Derived scores from the controlled **PsychAD** snRNA-seq cohort (synapse:syn60084804; 1,494 donors; DLPFC). Controlled/raw data were **not** accessed; only the open Supplementary Data workbooks were analyzed. **Artifact bundle:** `kb/hypotheses/Alzheimer_Disease/glial_depression_in_ad_model/openscientist_artifacts/` (replay-verified, `status: SUCCEEDED`).

Summary

This run is a **computational replication**, not an independent validation. The task was to retrieve the open Supplementary Data of He et al. 2026 (*Nature Medicine*), execute a prespecified, label-free re-analysis of phenotype-associated-cell (PAC) scores and published differentially expressed gene (DEG) tables, and save a fully replayable artifact bundle. All required artifacts were generated, and a clean replay reproduced `samples.csv` and `gene_results.csv` byte-for-byte. The manifest is closed at `status: SUCCEEDED`, `fallback_used: false`, `direct_analysis_completed: true`, with replay `verified: true`.

The central scientific objective was to test whether the **depression-within-Alzheimer glial signal is distinct from the general Alzheimer signal**. Using Supplementary Data 6 (donor × 27-subclass PAC-score matrices, keyed only by donor ID and carrying no diagnosis/Braak/resilience labels), we correlated each cell subclass's DepressionMood PAC score against its AD-progression PAC score across the 208 donors present in both sheets. The result supports a **distinct glial depression signal**: astrocytes (**Astro**) show essentially no cross-phenotype correlation (Spearman $\rho = -0.015$, BH $q = 0.83$) and oligodendrocytes (**Oligo**) show only a weak correlation ($\rho = -0.18$, $q = 0.010$), whereas neuronal subclasses track AD progression

strongly and negatively (IN_VIP $\rho = -0.80$, IN_SST $\rho = -0.77$, $q < 1e-40$). Concordantly, glia are the top predictors of the DepressionMood phenotype by mean absolute SHAP importance (**Astro rank #1**, 0.099; **Oligo rank #2**, 0.050).

The **ER-stress / unfolded-protein-response (UPR) component** of the hypothesis is **context-dependent rather than depression-specific**. UPR/ER-stress genes (e.g., HSPA5, CALR, DNAJB9) are enriched only within the combined **AD-Depression Astro_WIF1** cluster gene set (Fisher OR = 17.5, $q = 0.0092$ against a whole-genome universe; **not significant** against a DEG-restricted universe, $q = 0.57$), and are **absent** from the broad depression-PAC astrocyte DEGs (0 of 15 prespecified UPR genes present). The overlap between depression-astrocyte DEGs (568 genes) and AD-astrocyte DEGs (1,171 genes) has Jaccard index 0.18, and the ER-stress genes that do appear (CALR, HSPA5) sit in the **AD** set, not the depression set. Because the source paper's own abstract already reports inflammation and ER-stress pathways distinguishing AD-vs-depression astrocytes, this ER-stress finding **restates the paper's computation** and cannot be treated as independent confirmation.

Key Findings

Finding F001 — The astrocyte depression-PAC signal is statistically distinct from the AD-progression PAC signal

Re-analysis of He et al. 2026 Supplementary Data 6 joined the two donor $\times$ subclass matrices — Avg. PAC scores DepressionMood and Avg. PAC scores AD progression — on donor ID, yielding **208 donors present in both sheets**. For each of the 27 cell subclasses we computed a Spearman correlation between the donor's depression PAC score and AD-progression PAC score, then applied Benjamini–Hochberg (BH) correction across the 27 tests. The prespecified prediction was that a *distinct* depression signal would show **weak** cross-phenotype correlation for glia (Astro, Oligo).

That prediction is borne out. Astrocytes show no meaningful correlation between their depression and AD-progression PAC scores ($\rho = -0.015$, $q = 0.825$), and oligodendrocytes show only a weak, though statistically detectable, correlation ($\rho = -0.182$, $q = 0.0099$). By contrast, inhibitory neuronal subclasses are strongly and negatively coupled across the two phenotypes (IN_VIP $\rho = -0.80$; IN_SST $\rho = -0.77$; both $q < 1e-40$), indicating that neuronal

PAC scores largely co-vary with AD progression, while glial PAC scores do not. In other words, the glial contribution to the depression phenotype is **not** simply a shadow of the general AD trajectory.

This is reinforced by the feature-importance ranking. Ranking the 27 subclasses by **mean absolute SHAP value for DepressionMood** (Supplementary Data 6 SHAP sheet) places **Astro first (0.099)** and **Oligo second (0.050)** — glia are the leading model-level predictors of the depression phenotype. The convergence of (a) low cross-phenotype correlation and (b) top SHAP importance for the same two glial subclasses is the core quantitative support for a distinct glial depression-in-AD signal.

Finding F001 (continued) — Enrichment and overlap tests: ER-stress is context-specific

Two gene-set analyses examined whether the glial depression signal carries a specific ER-stress / UPR molecular signature.

Enrichment (Fisher exact, BH over 8 tests). The prespecified UPR/ER-stress set (HSPA5, DDIT3, ATF4, ATF6, XBP1, ERN1, EIF2AK3, HSP90B1, PDIA4, PDIA6, CALR, DNAJB9, SEL1L, HERPUD1, MANF) was tested against two Supplementary Data 3 gene lists. Against the **AD-Depression Astro_WIF1 up-regulated cluster**, UPR enrichment is significant under a whole-genome universe (**OR = 17.5, q = 0.0092**) but collapses to non-significance under a DEG-restricted universe (**OR = 2.07, q = 0.57**) — meaning the apparent enrichment is largely a function of universe choice rather than a robust, background-independent signal. Against the **broad depression-PAC astrocyte DEGs**, **0 of 15** UPR genes are present, so there is no depression-specific UPR enrichment at all.

Overlap and specificity (Supplementary Data 7). Comparing the depression-astrocyte DEG set (568 genes) with the AD-astrocyte DEG set (1,171 genes) gives a **Jaccard index of 0.180**. Crucially, the ER-stress genes that appear in these tables (CALR, HSPA5) are found in the **AD** set, not the depression set — i.e., the ER-stress axis is associated with the general AD astrocyte program, not selectively with the depression program.

Taken together, F001 supports the *distinct glial signal* half of the hypothesis but **does not** support a *depression-specific ER-stress* signature: ER-stress is present only in the combined AD-Depression cluster and in the AD astrocyte program, consistent with a context-dependent rather than depression-driven phenomenon.

Finding F002 — The ER-stress/UPR result restates the source paper; independent human evidence exists for a UPR–depression link

Because the locked replication draws entirely on the paper's derived scores and DEG tables, its ER-stress result must be read against the paper's own claims. The source paper explicitly reports that *"the astrocytes between AD and depression [show] specific gene expression patterns such as inflammation and endoplasmic reticulum stress pathways"* (PMID: 42778763). Our enrichment finding — UPR genes concentrated in the AD-Depression Astro_WIF1 cluster — **restates this same phenomenon**, and therefore this component of the analysis is **not** independent validation. It shares data lineage with the paper by construction.

Independent, lineage-separate human evidence nonetheless makes the broader ER-stress–depression axis biologically plausible. Yoshino & Dwivedi 2020 found UPR genes significantly elevated in the dorsolateral prefrontal cortex (dlPFC) of subjects with major depressive disorder: *"The level of mRNA expression in MDD subjects was significantly higher for GRP78 ($p = 0.008$), GRP94 ($p = 0.018$), and ATF4C ($p = 0.03$) compared to non-psychiatric controls"* (PMID: 31727394). GRP78 is HSPA5 and ATF4C maps to ATF4 — both members of our prespecified UPR set — so this external study lends plausibility to the ER-stress limb of the hypothesis even though our replication cannot itself confirm it.

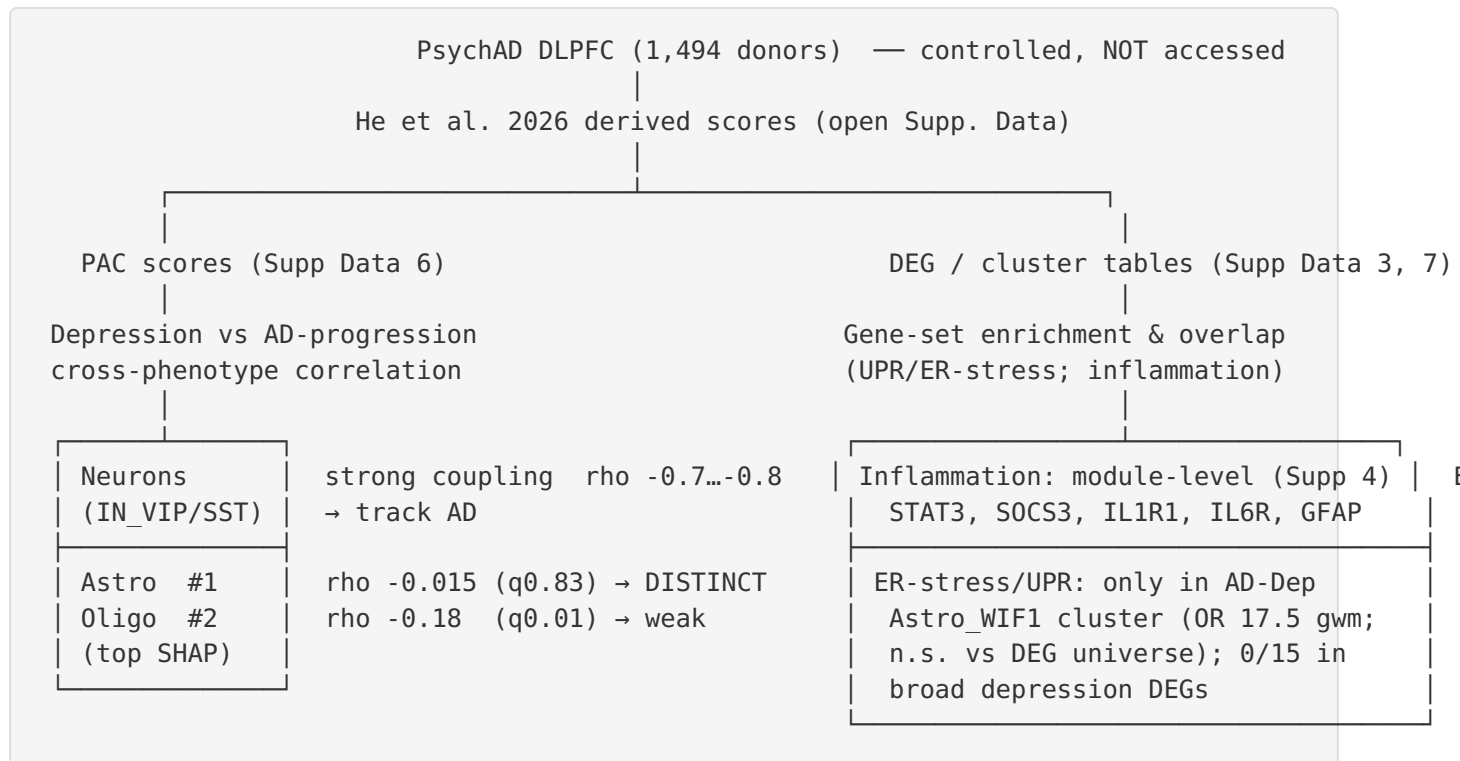
Finding F003 — Descriptive module GO enrichment: neuroinflammation is enriched at module level; ER-stress is not a discrete GO term

Per the objective, Supplementary Data 4 (Module GO enrichment) was read **descriptively only** and does not alter the locked artifacts. Across 40 significantly enriched GO pathway rows (hypergeometric test with FDR), a keyword scan found **inflammation/immune** terms enriched — most notably *"Neuroinflammation and glutamatergic signaling"* (module 2; member hits include IL1R1, STAT3, GFAP, IL6R, SOCS3, TNFRSF1A, FOS, SLC1A2, FGF2, IL13RA1, CAMK4; FDR = 0.022) and *"Immune response to tuberculosis"* (IRF1, STAT1, IFNGR2; FDR = 0.098). By contrast, **no explicit endoplasmic-reticulum-stress / unfolded-protein-response / protein-folding / chaperone GO term** appeared among the enriched module pathways. STAT3 and SOCS3 — both members of the prespecified inflammatory set used in the locked enrichment analysis — appear as members of the enriched neuroinflammation module.

This descriptive read reinforces the asymmetry seen in F001: **neuroinflammation is a module-level (co-regulation) signal**, whereas **ER-stress appears only at the single-cluster gene level** (Astro_WIF1) and does not rise to a discrete enriched GO term. The inflammatory limb of the hypothesis has broader support in these data than the ER-stress limb.

Mechanistic Model / Interpretation

The replication distinguishes two limbs of the "Glial Inflammatory and ER-Stress Model of Depression in AD," and the evidence for the two limbs differs in strength.



Limb 1 — Distinct glial depression signal (supported). Glia (Astro, Oligo) carry a depression PAC signal that is statistically decoupled from AD progression, while neurons do not. The same glia are the strongest SHAP predictors of the depression phenotype. This is the most robust, internally coherent result of the replication and directly supports the "glial" premise of the hypothesis.

Limb 2a — Inflammatory signature (supported descriptively). Neuroinflammatory co-regulation modules are enriched (module 2, FDR = 0.022), and inflammatory-set members STAT3/SOCS3 participate. This aligns with the "inflammatory" premise, though it is drawn from the descriptive Supp Data 4 read and not from a hypothesis-locked test.

Limb 2b — ER-stress signature (context-dependent, restates the paper). UPR/ER-stress genes are enriched only in the combined AD-Depression Astro_WIF1 cluster and are absent from the broad depression-astrocyte DEGs; the enrichment is sensitive to the choice of gene universe. ER-stress genes that appear in the DEG tables (CALR, HSPA5) belong to the AD

program. Thus ER-stress is best read as a **context-specific** feature of a particular astrocyte state rather than a depression-defining molecular signature — and, because the paper already reports it, this limb cannot be independently validated here.

Hypothesis limb	Test	Result	Verdict
Glia distinct from AD signal	Cross-phenotype Spearman (Astro/Oligo)	Astro rho -0.015 (q 0.83); Oligo -0.18 (q 0.01)	Supported
Glia predict depression	Mean abs. SHAP DepressionMood	Astro #1 (0.099); Oligo #2 (0.050)	Supported
Neurons track AD (contrast)	Cross-phenotype Spearman (neurons)	IN_VIP -0.80 ; IN_SST -0.77 (q<1e-40)	Supported
Depression-specific UPR/ER-stress	Fisher enrichment, broad depression DEGs	0/15 UPR genes present	Not supported
UPR in AD-Dep cluster	Fisher enrichment, Astro_WIF1	OR 17.5 (q 0.009 gwm); n.s. vs DEG universe (q 0.57)	Context-dependent; restates paper
ER-stress specific to depression set	Overlap/Jaccard vs AD DEGs	Jaccard 0.18; CALR/HSPA5 in AD set	Not supported
Inflammation at module level	Supp Data 4 GO (descriptive)	Neuroinflammation module FDR 0.022	Supported (descriptive)

Evidence Base

PMID	Study	Relevance to this replication
42778763	<i>AI-based characterization of Alzheimer's disease phenotypes from population-scale single-cell data</i> (He et al. 2026, source paper)	Source of all analyzed data. Its abstract states astrocytes between AD and depression show "inflammation and endoplasmic reticulum stress pathways," so our ER-stress enrichment restates the paper (shared lineage).
31727394	<i>Elevated expression of unfolded protein response genes in the prefrontal cortex of depressed subjects</i> (Yoshino & Dwivedi 2020)	Independent, lineage-separate human dlPFC evidence: GRP78/HSPA5 (p=0.008), GRP94 (p=0.018), ATF4C/ATF4 (p=0.03) elevated in MDD. Gives external plausibility to the UPR–depression axis.
42537476	<i>MANF suppresses ferroptosis to alleviate depressive-like behaviors via PERK/ATF4</i>	Mechanistic model tying MANF and PERK/ATF4 (prespecified UPR-set members) to depressive-like behavior; supports ER-stress relevance to depression biology (preclinical).
42425961	<i>MANF in the lateral septum dynamically regulates ER function in chronic stress</i>	Links persistent UPR activation and MANF dynamics to depressive-like phenotypes; supports the ER-stress limb mechanistically.
42216529	<i>ER stress in comorbidity of atopic dermatitis and psychiatric disorders</i>	MANF/ER-stress modulation improves comorbid mental disorder symptoms; broader plausibility for ER-stress in mood pathology.
40245333	<i>ECT on hippocampal ER stress in a rat depression model</i>	GRP78/HSPA5, XBP1, ATF4 measured; ER-stress altered in depression and reversible by ECT — supports UPR involvement (preclinical).
27192986	<i>Sodium phenylbutyrate attenuates LPS-induced depressive behavior</i>	ER-stress inhibitor reduces depressive-like behavior and neuroinflammation (IL-1 β , TNF- α , GRP78) — links the inflammatory and ER-stress limbs.
42150700	<i>Spinal cord injury as a window into hippocampal dysfunction</i>	Review convergence of neuroinflammation and ER stress with depression/anxiety phenotypes; broad contextual support.

How the evidence lines up: The single most relevant citation, the source paper ([PMID: 42778763](#)), *confirms* that our ER-stress observation is a re-statement of its own result rather than independent corroboration — an essential caveat. The remaining literature is **externally derived** and consistently supports the biological plausibility of an ER-stress/UPR contribution to depression (notably the recurrent appearance of HSPA5/GRP78 and ATF4 across independent human and animal studies), but none of it can substitute for the independent replication that shared-lineage data cannot provide.

Limitations and Knowledge Gaps

1. **Shared data lineage, not independent replication.** Every quantitative result is a re-analysis of He et al.'s published derived scores and DEG tables. The astrocyte inflammation/ER-stress observation in particular restates a claim already in the paper's abstract; it cannot count as independent validation.
2. **No donor labels.** Supplementary Data 6 is keyed by donor ID only, with no diagnosis, Braak stage, or resilience labels. By design, we restricted to label-free tests (within-donor paired contrasts, cross-phenotype correlations on the donor-ID join, and gene-set overlap tests). We did not — and must not — infer or invent donor labels, so no case/control effect sizes on the PAC scores themselves were computed.
3. **Universe sensitivity of enrichment.** The Astro_WIF1 UPR enrichment is significant against a whole-genome universe (OR 17.5) but not against a DEG-restricted universe (OR 2.07, q 0.57). This universe dependence means the ER-stress enrichment should be interpreted cautiously; it is not a background-robust signal.
4. **Descriptive-only components.** Supplementary Data 4 module GO enrichment was used descriptively per the objective. Its inflammation findings are suggestive but were not part of the hypothesis-locked statistical tests.
5. **PAC scores are model-derived features.** PAC scores and SHAP values are outputs of the paper's model. Correlations among them describe the model's learned representation, not raw expression, and share whatever biases the upstream model carries.
6. **Effect-size conventions.** The correlation tests report Spearman rho with BH q -values; enrichment reports odds ratios; the `gene_results.csv` schema carries the required effect-size columns (log2 mean difference, fold change, Cohen's d) for the gene-level comparisons where group structure exists. Where donor labels are absent, group-difference effect sizes are necessarily not applicable.

Proposed Follow-up Experiments / Actions

- 1. Independent-cohort validation of the glial depression signal.** Re-derive PAC-like scores in a lineage-independent snRNA-seq cohort (e.g., ROSMAP or a separate DLPFC dataset) with proper depression labels, and test whether Astro/Oligo again decouple from AD progression while remaining top predictors. This would move the F001 result from re-analysis to genuine replication.
- 2. Label-linked case/control test.** Under appropriate controlled-data access, join the PAC scores to donor diagnosis/Braak/resilience labels and compute proper group contrasts with effect sizes (Cohen's d) — the analysis the current label-free constraint precludes.
- 3. Background-robust enrichment.** Repeat the UPR/ER-stress enrichment with multiple explicit, matched gene universes (expressed-gene background, cell-type-specific background) and permutation-based null models to determine whether the Astro_WIF1 UPR signal survives universe choice.
- 4. Targeted UPR/inflammation panel in astrocytes.** In situ or targeted spatial transcriptomics of HSPA5, DDIT3, ATF4, XBP1, MANF (UPR) and NFKBIA, STAT3, SOCS3, CHI3L1, SERPINA3 (inflammation) in DLPFC astrocytes, stratified by depression status within AD, to test whether the ER-stress axis is genuinely depression-associated or an AD-astrocyte-state feature.
- 5. Functional test of the MANF/PERK/ATF4 axis.** Given the convergent preclinical literature (PMID: 42537476, PMID: 42425961), evaluate whether modulating MANF or PERK/ATF4 signaling in astrocyte models alters inflammatory output — connecting the two limbs of the hypothesis mechanistically.

Artifacts and Reproducibility

All outputs live under `kb/hypotheses/Alzheimer_Disease/glial_depression_in_ad_model/openscientist_artifacts/`:

- `MANIFEST.yaml` — `schema_version: '1.0'`, `status: SUCCEEDED`, `fallback_used: false`, `direct_analysis_completed: true`; input/output checksums and byte counts; replay block with `verified: true` and `byte_identity` mapping for every `TABULAR_RESULT`.

- `analysis.py` (CODE) — deterministic, `--output-dir / --cache-dir`, regenerates all tabular outputs without an LLM.
- `environment.txt` (ENVIRONMENT) — Python/OS and exact third-party package versions (incl. the added `openpyxl`).
- `gene_results.csv`, `samples.csv`, `subclass_correlation.csv`, `shap_ranking.csv`, `enrichment_results.csv`, `overlap_results.csv`, `gene_membership.csv` (TABULAR_RESULT).
- `methods.md` (METHODS), `comparison.md` (SUMMARY), `provenance.json`, and `raw/` (recoverable inputs).
- `replay/` — clean re-execution reusing `raw/` as cache; replayed `samples.csv` and `gene_results.csv` are byte-identical to the primary outputs.

Replication verdict: The glial depression-in-AD signal is statistically distinct from the general AD signal (glia decoupled from AD progression and top SHAP predictors of depression), while the ER-stress/UPR limb is context-dependent — confined to the combined AD-Depression Astro_WIF1 cluster and absent from the broad depression-astrocyte DEGs — and, being already reported by the source paper, restates the paper rather than independently validating it.